Video:

https://vimeo.com/82667821

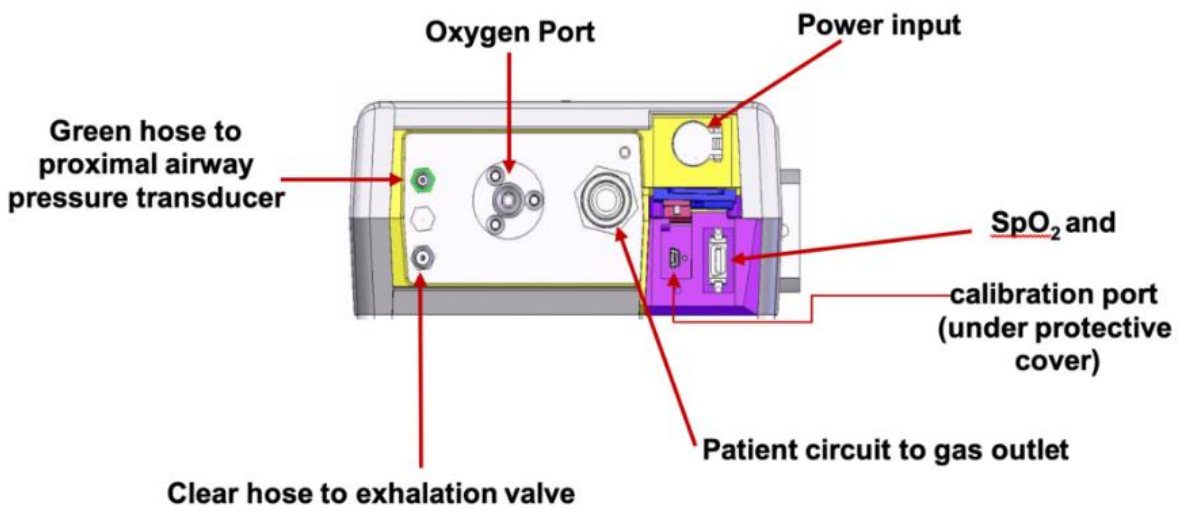
https://vimeo.com/82667822

Circuits:

- >25kg = Adult
- 5-25kg = Pediatric
- <5kg = bag patient

Set-up:

- Green to green (pressure transducer)
- Clear to silver (exhalation valve)
- Patient circuit to gas outlet



- In-line EtCO2 to patient circuit hook-up
- HME (green filter) to EtCO2
- Viral/Bacterial filter (white filter) to exhalation port

<u>Ventilator Setting Locations:</u>

FiO2

PIP:

- PEEP
- Pressure Support (PS): press and hold button for 2-3 sec and PS menu will pop up

Vt

BPM (RR):

- I:E ratio
- Press and hold for:
  - Rise Time (1= fast, 10 = slow)
  - Cycle off %

Mode:

- AC (P), AC (V), SIMV, CPAP
- PPV (positive pressure ventilation), NPPV (noninvasive PPV)
- Apnea back up in NPPV: press and hold button to pop up menu with BPM, PIP target, I:E

Menu Button:

- Alarm Config- turn off/cancel alarms (automatically turn back on next time vent turned on)
- Powerup Settings- DO NOT TOUCH THIS (changes initial vent settings on start up)
- Pulse Oximeter- allows you to connect pulse ox to the vent, we don’t do this
- Trigger level- adjust assisted breath trigger
- O2 Reservoir- we don’t use this (tell ventilator how much O2 you have)

- Unit Info
- Contrast- adjusts display contrast

Alarm Info

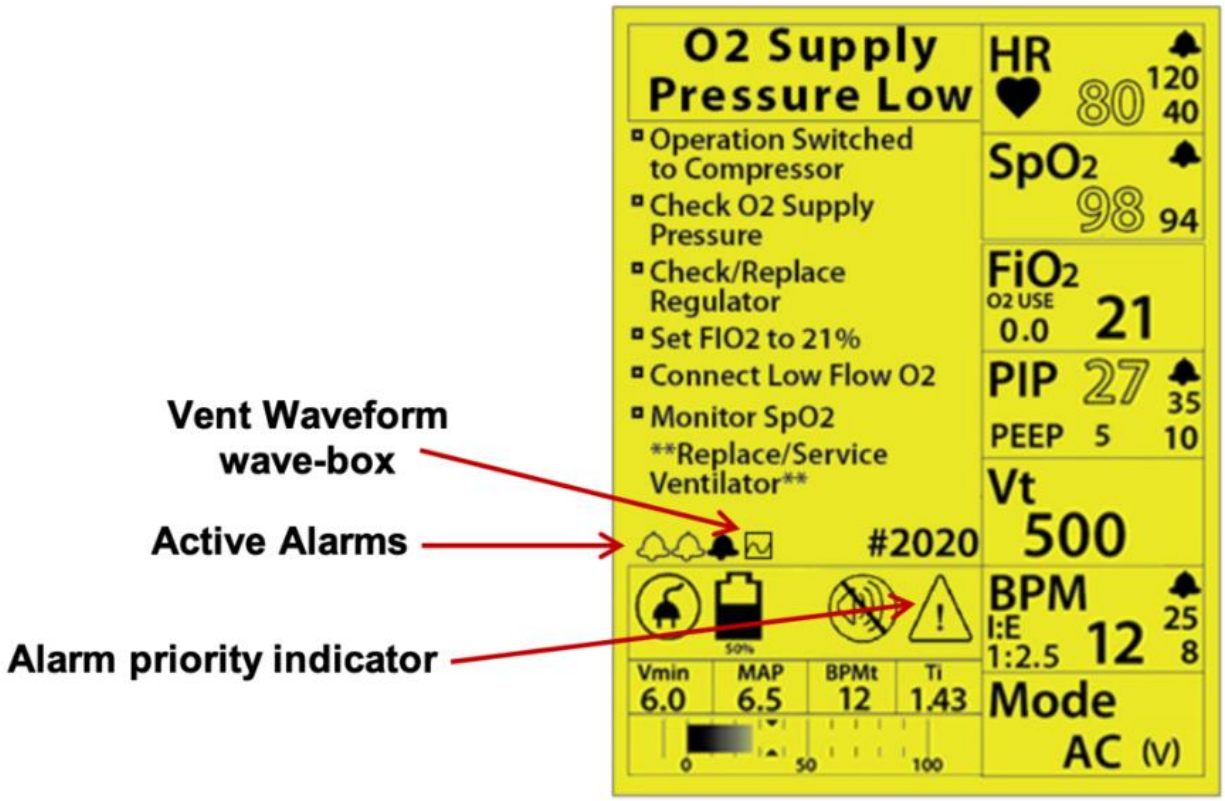
- Automatically appears in upper left when alarm active
- Recommends corrective actions
- To switch between multiple alarms, scroll over clear bell and select (current alarm will be a solid bell)

Alarm Priority Indicator: Triangle to left of BPM

- No ! = low priority
- ! = medium priority
- !! = high priority (you cannot mute these)

Vent Waveform (pressure/time or volume/time)

- Automatically appears on upper left
- If alarm text blocking waveform, scroll over bells to wave-box then select



VENTILATOR ALARMS:

PEEP Leak

- What: Measured PEEP drops by ≥ 2
- Common Cause: Leak in circuit, exhalation valve, patient airway
- Fix: ensure all connections are tight

Incomplete Exhalation

- What: continued expiratory flow during initiation of next breath
- Common Cause:
  - Patient: bronchoconstriction, ARDS (look at your patient then vent scalar)
  - System: inline Nebs, HME, suction catheters (increase expiratory resistance)
- Fix:

- If air trapping, increase I:E ratio to allow full exhalation
- If system, can ignore/turn alarm off (caution if patient has any risk factors for true incomplete exhalation)

Gas Intake Fault

- What: fresh gas/emergency air inlet (upper right side) blocked
- Common Cause: external blockage
- Fix: remove blockage

Inspiratory Demand Not Met

- What: end inspiratory pressure < -1.0 (more negative)
- Common Cause: deep inspiration by patient (air hunger, neuro breathing pattern, compensating for metabolic acidosis, etc.)
- Fix:
  - If fighting vent/neuro breathing, consider sedation
  - If air hunger, increase flow rate via decreased I time or rise time

SETTING UP BI-LEVEL VENTILATION

Ventilator Settings:

- Set Mode to CPAP
- Change PPV to NPPV in Mode box (press Mode button twice to highlight PPV)
- Set PEEP
- Press and hold PIP button for 2-3 sec to open Pressure Support Menu
- Set Pressure Support
  - This is your delta pressure!!
  - If PEEP increases PIP automatically increases to keep constant delta pressure/ventilatory support

Troubleshooting

- Patient cannot breath in for as long as they want to —> Increase cycle off %
  - Due to air leak, vent switches to exhalation too early

Source:

Air Care Equipment — Taming the SRU